

Designing a Usable Framework for Diverse Users in Synthetic Human Action Data Generation

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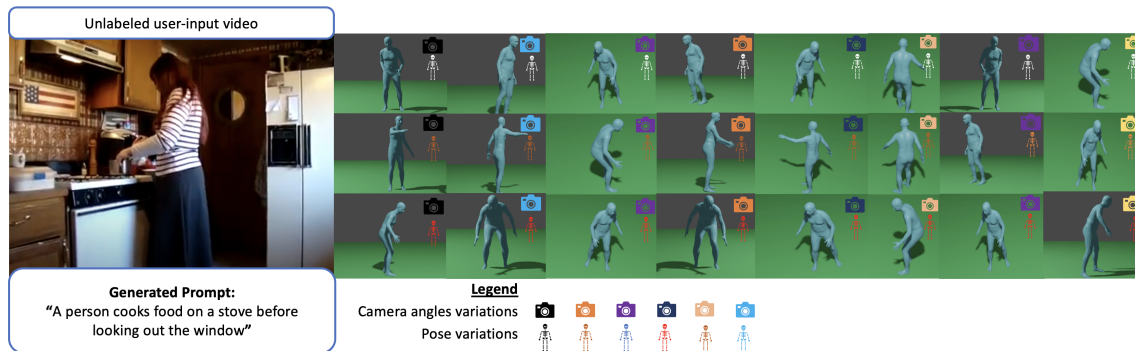


Figure 1: Examples of synthetic data (right grid) generated with SynthDa2 from an input video (left image, resolution as per source). Each output in the grid represents a complete video sequence, with new camera angle and pose variations (see Legend).

Abstract

This paper introduces SynthDa2, a synthetic data generation framework aimed at addressing data scarcity for training video-based machine learning models. Grounded in an initial user study (n=84), SynthDa2 includes both an API and a UI to accommodate technical and non-technical users. The framework leverages generative AI techniques and domain randomization to create diverse, user-customized synthetic video datasets. Case study experiments on dataset permutations demonstrate the feasibility of SynthDa2 in assessing composition impacts. While the initial user study confirmed camera angles as a popular key variation factor, experiments reveal they are insufficient alone for desired outcomes, highlighting the

need for further research. A subsequent hands-on user study (n=8) further validates SynthDa2's usability across varied users.

CCS Concepts

• Computing methodologies → Graphics systems and interfaces; Machine learning; Computer vision.

Keywords

synthetic data generation, action detection, user study, human-computer interaction

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1 Introduction

Synthetic data generation (SDG) for human action has advanced in addressing data scarcity, but often overlooks user needs. This



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paper introduces SynthDa2, a user-focused SDG framework using generative AI to create diverse datasets that address these needs. To address data scarcity, SDG tools must be practical and user-friendly [Ramesh et al. 2023]. Integrating user-centric design is essential [Abrams et al. 2004], but a gap exists between SDG's potential and user experience due to limited insights into end-user needs and lack of formal validation of variation factors. Understanding these factors is key to improving adoption and utility. Previous studies have explored SDG, but gaps remain in validating the impact of variation factors on model training. (Sec. 2). This paper hence seeks to address these gaps by answering the following research question (RQ):

Can we design a usable SDG framework that effectively generates diverse synthetic data, allowing meaningful assessment of different compositions of variation factors on model training?

Following user-centered design principles [Abrams et al. 2004], insights from an initial user study (US1) with 84 participants guided the development of SynthDa2, incorporating both a user interface (UI) and an application programming interface (API) to ensure usability for both non-technical and technical users. Use case experiments demonstrate SynthDa2's ability to generate diverse synthetic datasets for meaningful data composition assessments. Usability was further validated in a follow-up user study (US2), where participants confirmed its effectiveness and ease of use.

The key contributions of this paper are: (1) Analyzing user needs and perceptions regarding SDG variation factors through US1; (2) Presenting the SynthDa2 framework, a modular LLM-based system accessible via both UI and API; (3) Conducting use case experiments to demonstrate technical feasibility; and (4) Validating SynthDa2's usability through US2.

2 Related Work

Existing SDG methods rely on GANs [Clark et al. 2019], 3D authoring [Roitberg et al. 2021], and domain randomization [Tremblay et al. 2018], but face accessibility issues due to specialized skills and high costs. Domain randomization, still in its early stages, lacks standardized frameworks, limiting its use.

Current literature generally agrees on the essential variation factors for generating datasets [Liu et al. 2019; Roitberg et al. 2021], but formal validation of these factors in such datasets [Liu et al. 2019] is lacking. There is a significant gap in understanding and validating common SDG variation factors for both trained and untrained users. Existing SDG tools are cumbersome and lack user-centric designs, hindering widespread adoption. Additionally, there is insufficient literature on the usability of current SDG tools.

Generative AI tools, (e.g., LLMs) [Lin et al. 2023; OpenAI 2024], enhance SDG frameworks' flexibility. The SynthDa2 framework utilizes these transformer-based models to enhance SDG processes, increasing synthetic dataset diversity with novel pose variations.

SynthDa [Rajendran et al. 2023] is an SDG pipeline to generate videos by extracting pose data from input videos. However, it had fixed camera angles, required Omniverse knowledge, and lacked pose variation and generative AI support, limiting it to pose reproduction. This paper tackles these issues by incorporating US1 insights into SynthDa2, creating a modular, user-friendly design. The framework supports configurable variation factors like pose

and camera angles (Fig. 1), reducing the learning curve for beginners and emphasizing the role of variation factors. An included API enhances flexibility for developers. A follow-up study (US2) with 8 participants demonstrate SynthDa2's potential usability and practicality for real-world applications.

3 SynthDa2

We identified a gap in the literature on validating common SDG variation factors and understanding user needs in SDG tools (Sec. 2). To address this (RQ), the SynthDa2 framework was developed.

3.1 Initial User Study (US1)

Our online qualitative survey with 84 participants (aged 21–65, $M=29.2$, $SD=10.9$) gathered design insights for SynthDa2 by understanding user needs. The cohort included 16 industry practitioners, 16 researchers, and 52 novices. Content analysis [Krippendorff 2019] was used to derive themes and validate variation factors for SDG.

3.1.1 Procedure. Recruitment emails were sent via mailing lists. The survey collected data on perceptions of SDG requirements, opinions on SDG tools, including a visual demo of an existing tool [Rajendran et al. 2023] to standardize the tool's visualization for all participants. The survey questions, conceived based on our RQ, targeted understanding perceptions of perceived usefulness of SDG tools and user needs. The study received IRB approval. Three researchers standardized coding through regular meetings to avoid bias. Two coded 80% of the data, with the remaining 20% handled by two independent coders (Cohen's Kappa, $\kappa = 0.78$). Regular discussions, led by the first author, ensured consistency. Details on US1 are in the supplementary material.

3.1.2 Theme 1: Camera angles and pose variation perceived as most important for SDG (T1). There is a lack of formal validation to identify essential variation factors for SDG (Sec. 2). In this study, participants expressed a strong need for 'VARIATION' (51/84) and interest in 'CONFIG-AVATAR-MODELS' (10/84). The most mentioned variation factors were 'CAMERA-ANGLES' (39/84) and 'POSE-MANIPULATIONS' (39/84).

P18: "(Varying) camera perspectives and angles; ... crucial to ensuring the model can adapt to different viewpoints, promoting robustness." [CAMERA-ANGLES]

P16: "It would be very useful if I could have a large library of animations that can be brought into the simulation and randomized re-targeting on different digital human assets." [POSE-MANIPULATIONS]

3.1.3 Discussion. There was minimal emphasis on visual avatar variations, such as 'SKIN-COLOR-REPRESENTATION' (3/84) and 'BODY-SIZES' (2/84). This theme provides new insights into user needs, contrasting with prior works [Roitberg et al. 2021], which included appearance variation factors in SDG [Liu et al. 2019]. Our findings indicate a preference for camera angle and pose manipulations over appearance variations. These insights guide the development and design of SynthDa2, highlighting the need to address user preferences such as options for variations in camera angles.

3.1.4 Theme 2: Usefulness in SDG Tools (T2). SynthDa [Rajendran et al. 2023], an existing SDG tool, was demonstrated

through a video in US1 for visualization and feedback (Sec. 3.1.1). Post-demonstration, 63% of participants (53/84) deemed SDG tools 'GENERALLY-USEFUL', emphasizing the necessity for low learning barriers with 'BEGINNER-LEVEL-EXPERIENCE' (44/84). SDG tool design should support skill transfer for wider usability, as highlighted by 'CROSS-EXPERIENCE-COMPATIBILITY' (47/84).

P51: "(Such tools can be) introduced to people who are new to generative AI and techniques." [BEGINNER-LEVEL-EXPERIENCE][CROSS-EXPERIENCE-COMPATIBILITY]

3.1.5 Discussion. Existing SDG tools often face usability challenges, requiring specialized skills such as machine learning expertise for model-based methods or proficiency with 3D authoring tools [Roitberg et al. 2021]. 'CROSS-EXPERIENCE-COMPATIBILITY' highlights the importance of skill transfer to improve practicality in SDG tool development. Responses show a high demand for usable, extensible tools to ensure real-world application [Ramesh et al. 2023]. Insights from T1 and T2 guided SynthDa2's framework design.

3.2 SynthDa2: Architecture & Design

SynthDa2 generates synthetic video data of human actions by integrating user-input videos with generative AI techniques (Fig. 2). The user-centered design, guided by US1 and based on design principles [Abrams et al. 2004], supports both API and UI for enhanced usability for technical and non-technical users (T2). SynthDa2 simplifies the input process using generative AI (T2) and enables users to include variation factors during data generation (T1). The framework, shown in Fig. 2, follows a modular design with two main components: the prompt generator and the video synthesizer, shaped by US1 insights (T1 & T2) to ensure a user-friendly experience.

3.2.1 Prompt Generator. From a user-provided video input s (Fig. 2), a *prompt* is extracted using motion captioning [Lin et al. 2023]. Users can select LLMs such as Video-LLaVA, ChatGPT, or incorporate their own models [Lin et al. 2023; OpenAI 2024]. Users configure variation factors, such as the number of pose variations per prompt and specific camera angles, denoted by $U = \{u_1, \dots, u_n\}$. The generated prompt is combined with user parameters, e.g., camera angles. The prompt generator is designed with a focus on essential variation factors, such as pose manipulations and camera angles, allowing users to easily configure these elements (T1). The process is further simplified by automatically extracting prompts from the input video, enhancing usability and minimizing the technical expertise required (T2) (Fig. 2).

3.2.2 Video Synthesizer. This component comprises three sub-components (below). The video synthesizer is designed to be modular, enabling users to swap sub-components for specific use cases or integrate the latest advancements into their workflows (T2).

3.2.3 Pose Generation. The prompt, *PROMPT*, is used to generate 3D pose sequences, $P = (p_1, \dots, p_T)$ (Fig. 2). SynthDa2 uses generative techniques [Lin et al. 2023] in a two-step approach: first, it applies a learned motion length distribution conditioned on input texts; then, a temporal variation autoencoder generates diverse motions for sampled lengths, using 6D continuous rotation for

pose data mapped to a 3D model. 'POSE-MANIPULATIONS' was validated (T1) as a key variation factor.

3.2.4 Domain Randomization. User-defined domain randomization [Tremblay et al. 2018], F , shifts focus from photorealism to targeted alteration of factors in videos. Focusing on selected variation factors enriches scene data, *scene*, thereby increasing dataset diversity. Variation factors (T1), such as 'CAMERA-ANGLES' (39/84) exemplify users' need for configurable variation factors in the domain randomization sub-component of the framework (Fig. 2).

3.2.5 Scene Construction. This integrates scene data with generated human pose data and variation factors to output videos (Fig. 2). This integration aligns synthetic human poses with dynamically configured variation factors, generating diverse synthetic videos V . This simulates diverse variations of scenarios, thereby improving model generalization to real-world actions and scenarios.

3.3 Use Case Experiments

To demonstrate data scarcity, a subset of the Charades dataset with 51 action classes was used as the baseline for 127 experiments. We trained a CV model [Dai et al. 2022] using datasets from SynthDa2 and real Charades data [Sigurdsson et al. 2016]. Standardized Charades video descriptors ensured consistency (Sec. 3.2.1). Variation factors from T1 and previous works [Liu et al. 2019] were configured. Datasets with 3 camera angles (left' [L], upper right' [UR], front' [F]) and 2 pose variations (A' and 'B') provided diverse configurations for performance evaluation.

3.3.1 Use Case Experiments Analysis & Discussion. Our composition analysis experiments offer a focused investigation, contrasting with benchmark studies like NTU RGB+D [Liu et al. 2019], which use numerous camera angles. The experiment's unexpected results showed decreased model performance of up to -10.04% (Fig. 3) suggesting that excessive camera angle variations may not be the most effective.. This also provides potential for further research as it appears to contradict US1's finding on the desire for camera angles diversity. This highlights that actual model performance may not align with predicted performance of popular variation factors. This suggests that it is important for users to evaluate and identify optimal variation factor compositions for their unique use cases. Our use case experiment here highlights how SynthDa2 plays an important role for users to effectively perform a meaningful assessment of variation factors for model training (RQ).

3.4 Follow-Up User Study (US2)

To validate the usability of SynthDa2 from the previous study (Sec. 3.1) we conducted a follow-up hands-on user study. The study, involving 8 participants (3 industry practitioners and 4 untrained users), yielded promising insights into the usability and potential applications of SynthDa2. Participants engaged in hands-on trials with SynthDa2 to generate synthetic data from a video of their choice and completed a subsequent survey. The tool was generally found to be user-friendly (M:6.33/7), with perceived usefulness scores (M:5.67/7) and most participants believed it could be quickly learned by others (M:5.67/7). Via content analysis [Krippendorff 2019] on participants' qualitative responses, we see that the ability to generate animations based on reference videos and prompts was

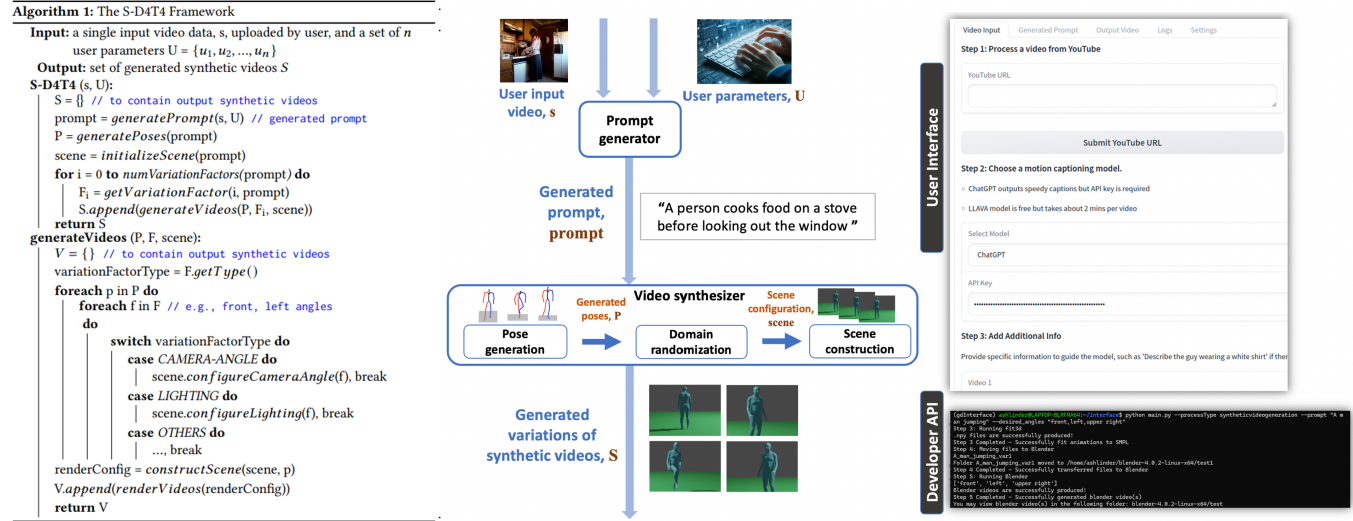


Figure 2: The SynthDa2 system diagram has two main components: the prompt generator and the video synthesizer. The prompt generator extracts prompts from the input video s , which the video synthesizer processes. It includes pose generation, domain randomization, and scene construction.

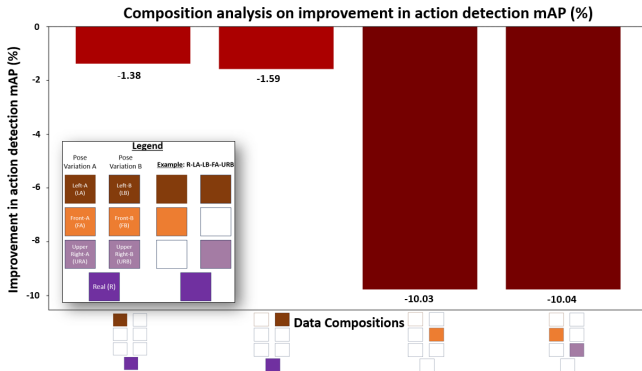


Figure 3: Use case experiments show up to a 10.04% (FA-URB) drop in model performance with varied camera angles. A total of 127 experiments were conducted using mixtures of generated data and Charades [Sigurdsson et al. 2016].

particularly well-received, across various aspects, suggesting potential applications in machine learning and data-related projects.

3.4.1 Discussion. The survey scores highlight SynthDa2's potential for future SDG design, though the small sample size of 8 limits statistical power, requiring further studies. Qualitatively, participants saw applications in AI research, game development, and video analysis, noting its flexibility over prior works [Roitberg et al. 2021]. The tool's animation generation from videos and prompts was well-received, showing improvements over other methods [Tremblay et al. 2018]. Suggested improvements include processing speed, video selection guidance, and UE Metahuman integration. Overall, SynthDa2 shows promise as a versatile SDG tool.

4 Conclusion

This paper introduces the SynthDa2 framework and design, informed by two user studies. The first identified key variation factors and usability guidelines, while the second confirmed usability and potential across diverse users. A use case addressing data scarcity was demonstrated through experiments. Future work should focus on improving UI guidance for non-technical users, expanding US2 to a larger cohort, and exploring additional use cases.

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